DSA 8020 R Lab 1: Simple Linear Regression

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Contents

Leaning Tower of Pisa

The dataset `PisaTower.csv` provides annual measurements of the lean (the difference between where a point on the tower would be if the tower were straight and where it actually is) from 1975 to 1987. We would like to characterize lean over time by fitting a simple linear regression.

Load the dataset

Code:

```
getwd()
```

```
## [1] "C:/Users/sar-home/Clemson/Spring_2025_Semester_2/DSA-8020/R_Labs"
```

```
PisaTower <- read.csv("datasets/PisaTower.csv")
head(PisaTower)
```

```
##     lean year
## 1 2.9642 1975
## 2 2.9644 1976
## 3 2.9656 1977
## 4 2.9667 1978
## 5 2.9673 1979
## 6 2.9688 1980
```

Descriptive analysis

Numerical summary

Provide some numerical summaries to describe the response and the predictor variables, respectively, as well as their relationship.

Code:

```
str(PisaTower)
```

```
## 'data.frame':    13 obs. of  2 variables:
##  $ lean: num  2.96 2.96 2.97 2.97 2.97 ...
##  $ year: int  1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 ...
```

```
summary(PisaTower$lean)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.964   2.967   2.970   2.969   2.972   2.976
```

```
summary(PisaTower$year)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1975    1978    1981    1981    1984    1987
```

```
summary_lean <- PisaTower %>%
  summarise(
    mean_lean = mean(lean, na.rm = TRUE),
    median_lean = median(lean, na.rm = TRUE),
    sd_lean = sd(lean, na.rm = TRUE),
    min_lean = min(lean, na.rm = TRUE),
    max_lean = max(lean, na.rm = TRUE)
  )

summary_lean
```

```
##   mean_lean median_lean     sd_lean min_lean max_lean
## 1  2.969369      2.9696 0.003651115   2.9642   2.9757
```

```
summary_year <- PisaTower %>%
  summarise(
    mean_year = mean(year, na.rm = TRUE),
    median_year = median(year, na.rm = TRUE),
    sd_year = sd(year, na.rm = TRUE),
    min_year = min(year, na.rm = TRUE),
    max_year = max(year, na.rm = TRUE)
  )

summary_year
```

```
##   mean_year median_year sd_year min_year max_year
## 1      1981        1981 3.89444     1975     1987
```

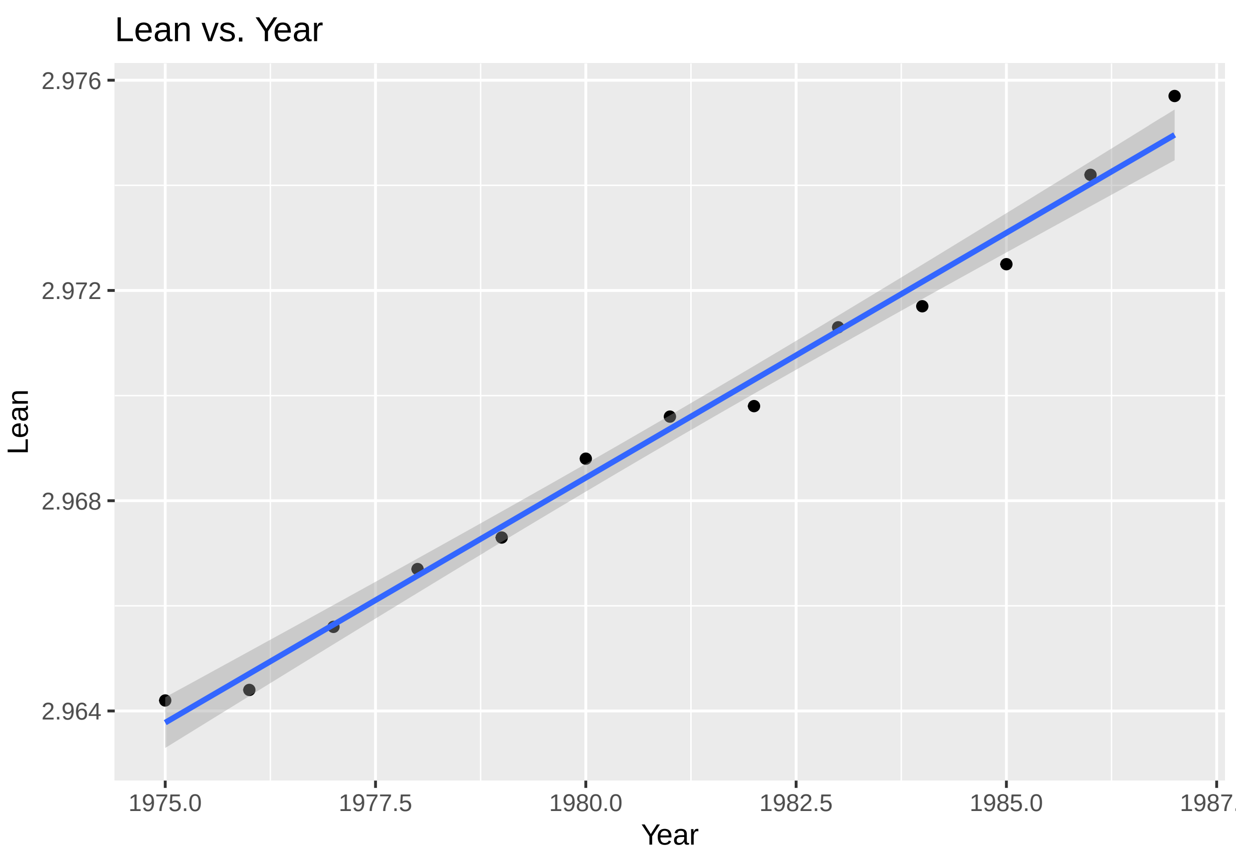
Graphical summary

Provide graphical summaries through plots to describe the response and predictor variables, respectively, as well as their relationship.

Code:

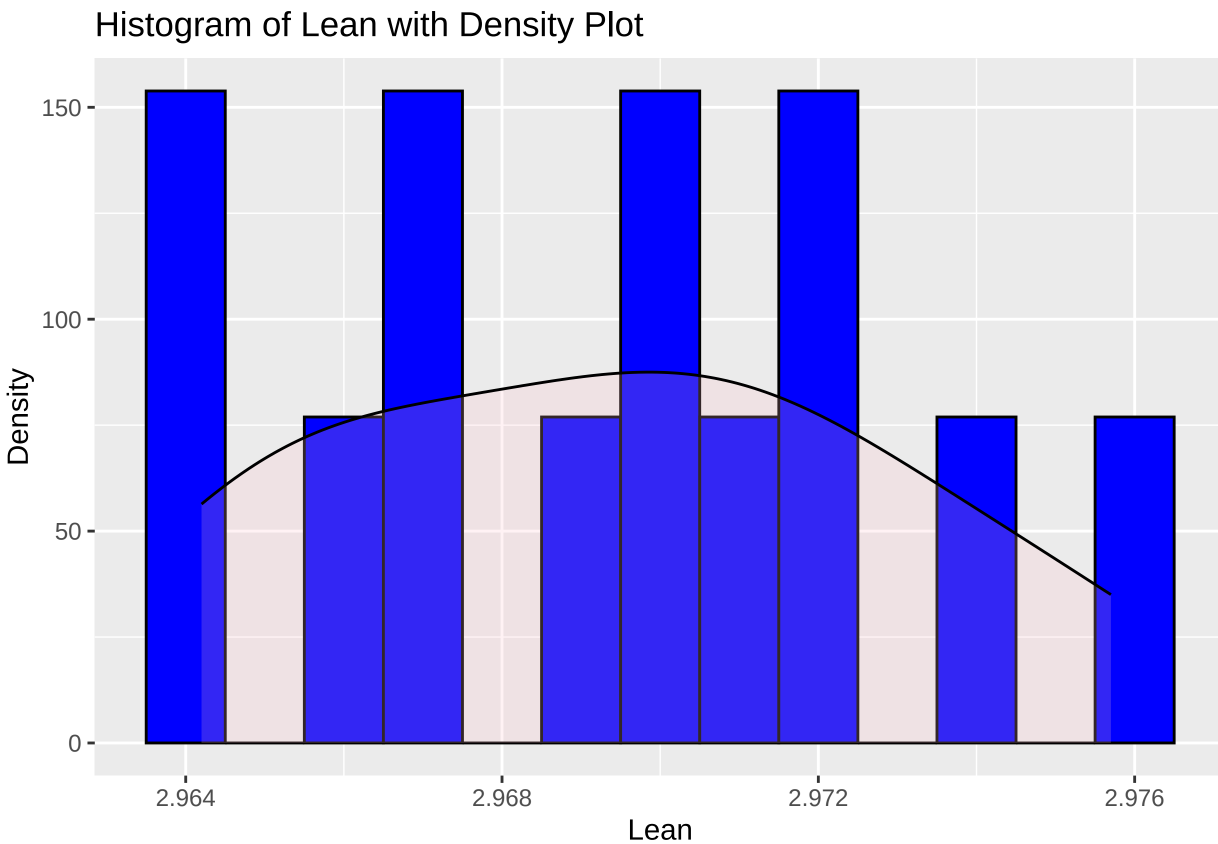
```
ggplot(PisaTower, aes(x = year, y = lean)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Lean vs. Year", x = "Year", y = "Lean")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
ggplot(PisaTower, aes(x = lean)) +
  geom_histogram(aes(y = ..density..), binwidth = 0.001, fill = "blue", color = "black") +
  geom_density(alpha = .2, fill = "pink") + # Adding some transparency to the density plot for better v
  labs(title = "Histogram of Lean with Density Plot", x = "Lean", y = "Density")
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Question: Describe the direction, strength, and the form of the relationship.

Answer:

For our scatter plot:

> The graph indicates that generally the lean values are increasing with time (year). The data points' general upward trajectory when plotted against time suggests a positive trend. The relationship heading has a positive feel to it, as there is one overall increase that exists in lean and year. If the points are close together with a little scatter, the relationship is strong, as we can see from the spread of the data points. On the other hand, points that are scattered or spread out suggest a less strong relationship. The level of correlation is not something we can quantify precisely without various statistical measures like the correlation coefficient. The linear regression line drawn through the data points shows that the form of relationship is likely linear. A straight line or smooth curve would suggest that the lean increases steadily through time.

For our histogram:

> This histogram most likely shows the distribution of lean values over years. This shows the frequency of occurrence of each range of lean values in your data. You can determine the variation in data and normality of your data based on the shape and spread of The histogram. For example, a roughly symmetric bell curve (e.g., Normal) indicates a strong linear connection. One may also speak of an independence assumption. What breaks your limits If you see some peaks or modal values in the histogram it may suggest the existence of some clusters or a common range of lean values.

Simple linear regression

1. Identify the response variable, the predictor variable, and the sample size.

Answer:

Response Variable: - Lean

Predictor Variable: - Year

Sample Size: - n = 13

2. Fit a simple linear regression.

Code:

```
PisaTower <- read.csv("datasets/PisaTower.csv")
head(PisaTower)
```

```
##     lean year
## 1 2.9642 1975
## 2 2.9644 1976
## 3 2.9656 1977
## 4 2.9667 1978
## 5 2.9673 1979
## 6 2.9688 1980
```

```
nrow(PisaTower)
```

```
## [1] 13
```

```
lm <- lm(lean ~ year, data = PisaTower)
summary(lm)
```

```
##
## Call:
## lm(formula = lean ~ year, data = PisaTower)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -5.967e-04 -3.099e-04  6.703e-05  2.308e-04  7.396e-04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.123e+00  6.139e-02   18.30 1.39e-09 ***
## year        9.319e-04  3.099e-05   30.07 6.50e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0004181 on 11 degrees of freedom
## Multiple R-squared:  0.988,  Adjusted R-squared:  0.9869
## F-statistic: 904.1 on 1 and 11 DF,  p-value: 6.503e-12
```

3. Write down the fitted linear regression model.

Answer:

$\hat{lean}$ = 1.123 + 0.0009319 * year

4. What is $\hat{\sigma}$, the estimate of σ?

Answer:

$\hat{\sigma}$ = 0.0004181 where $\hat{\sigma}$ is Residual standard error.

5. Find a 95% confidence interval for β_1.

Code:

```
confint(lm, level = 0.95)
```

```
##                    2.5 %     97.5 %
## (Intercept) 0.9882109317 1.25846599
## year        0.0008636565 0.00100008
```

```
beta1_hat <- lm$coefficients[2]  # slope estimate
se_beta1_hat <- summary(lm)[["coefficients"]][2, 2] # standard error

n <- nrow(PisaTower) # count observations
alpha <- 0.05 # significance for .95 confidence interval

t <- qt(1 - alpha / 2, n - 2)

# Confidence Interval
beta1_ci <- beta1_hat + c(-1, 1) * t * se_beta1_hat
beta1_ci
```

```
## [1] 0.0008636565 0.0010000798
```

6. Test the following hypothesis: $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$ with $\alpha = 0.05$

Answer:

Reject H_0. The p-value is 6.503367e-12, far smaller than $\alpha = 0.05$.

```
beta1_hat <- lm$coefficients[2]
se_beta1_hat <- summary(lm)[["coefficients"]][2, 2]

t_stat <- beta1_hat / se_beta1_hat
n <- nrow(PisaTower)
df <- n - 2

# P-value two tailed test
p_val <- 2 * pt(-abs(t_stat), df)

t_stat
```

```
##     year
## 30.06858
```

```
p_val
```

```
##         year
## 6.503367e-12
```

7. Construct a 90% confidence interval for E[`lean`] in year 1984.

Code:

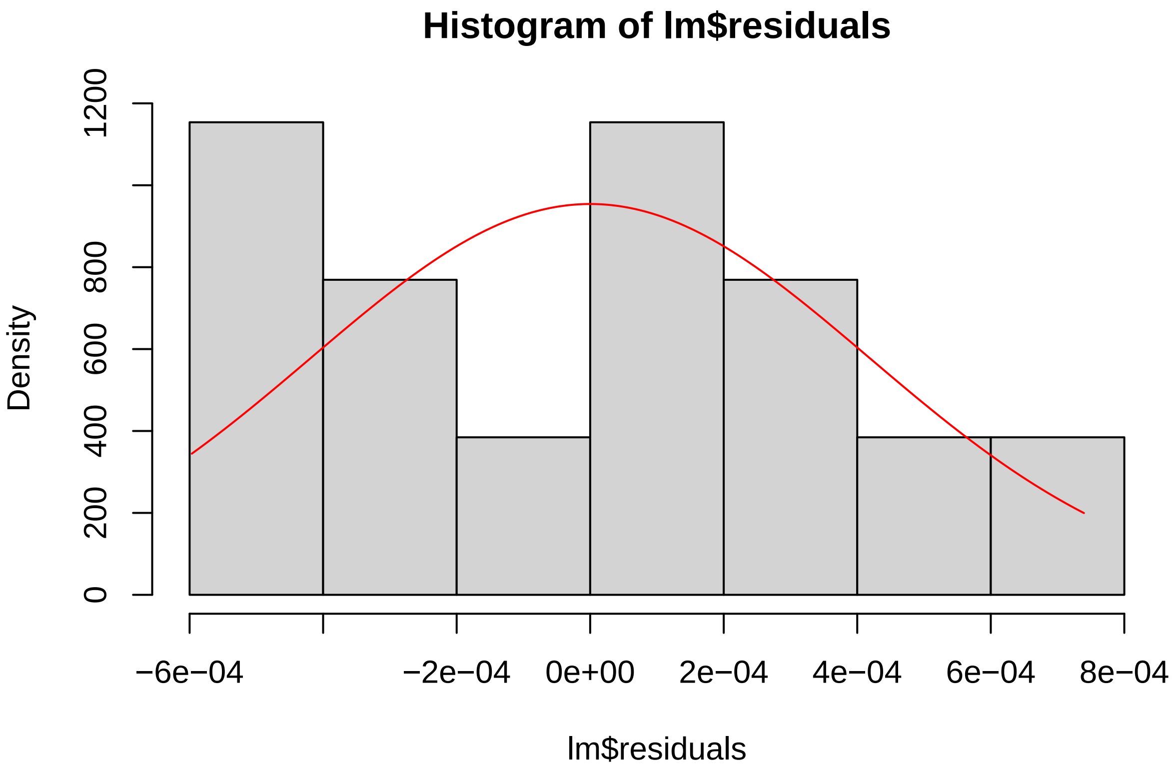
```
data_1984 <- data.frame(year = 1984)
predict(lm, newdata = data_1984, interval = "confidence", level = 0.90)
```

```
##        fit      lwr      upr
## 1 2.972165 2.971898 2.972432
```

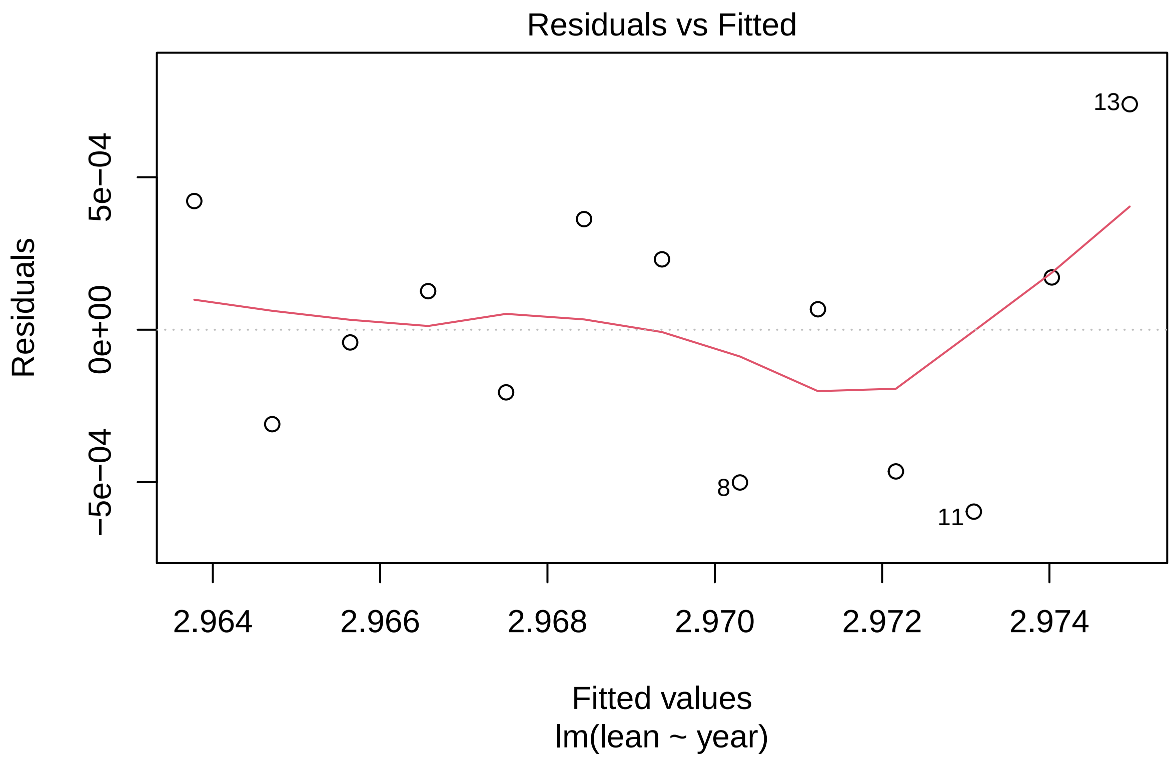
8. Use residuals to check model assumptions.

Code:

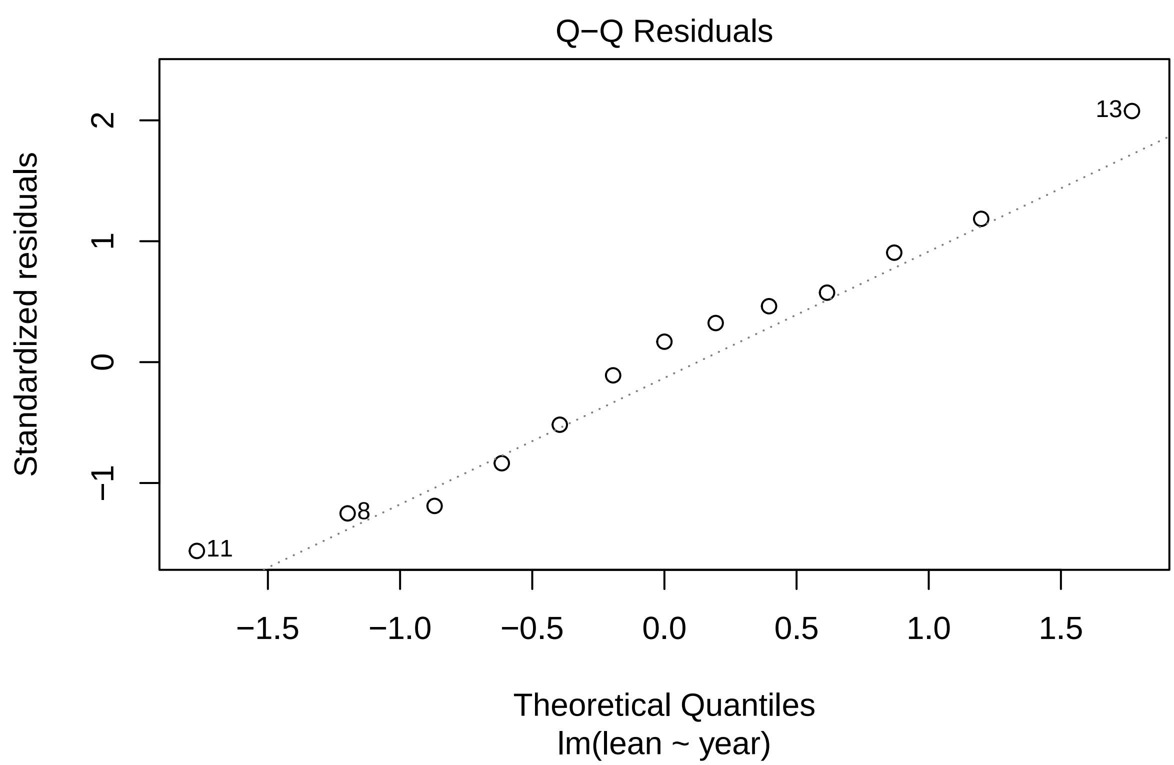
```r
hist(lm$residuals, 5, prob = T)
sigma <- summary(lm)$sigma

rg <- range(lm$residuals)
xg <- seq(rg[1], rg[2], len = 500)
yg <- dnorm(xg, mean = 0, sd = sigma)
lines(xg, yg, col = "red")
```



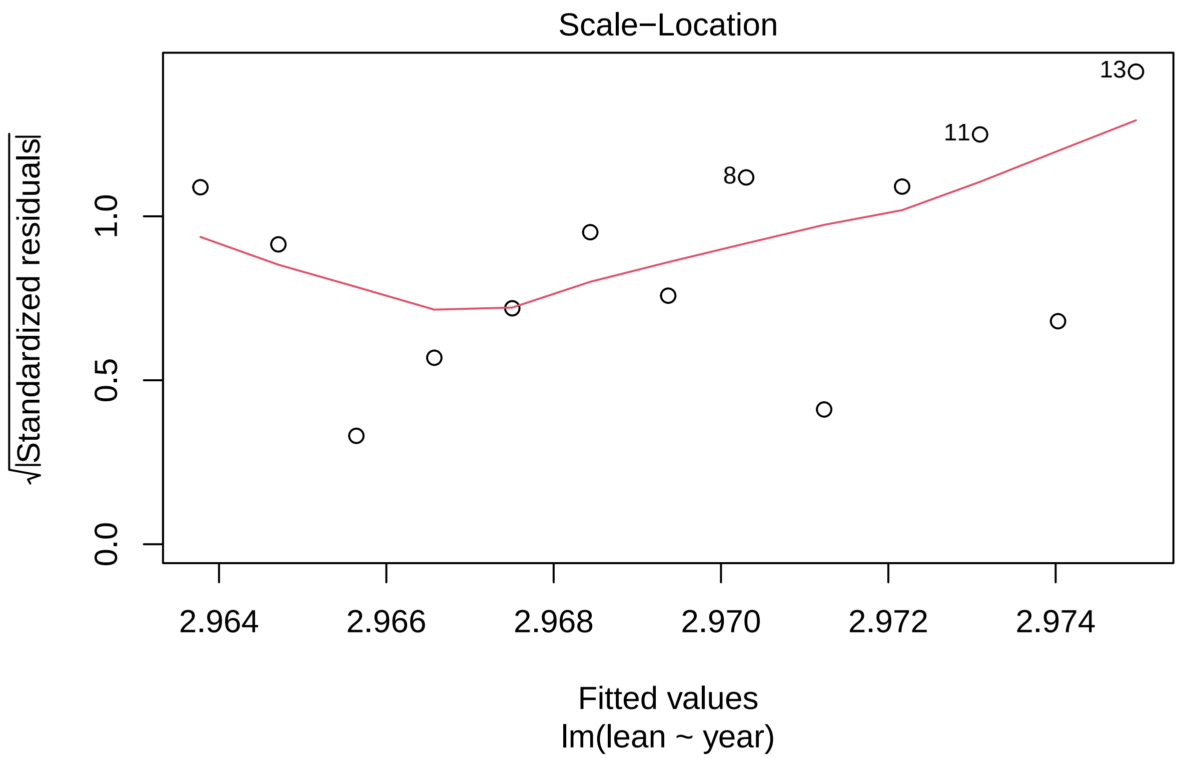

```r
plot(lm, which = 1)
```




```r
plot(lm, which = 2)
```




```r
plot(lm, which = 3)
```



Answer:

The residuals are not perfectly normal, which will affect statistical inferences (i.e. hypothesis tests, confidence intervals).

A. Linearity :

- *(Res VS Fitted Plot)* The residuals show a curved pattern rather than random scatter, which suggests non-linearity. This means the model may not fully capture the relationship between the predictors and the response variable.

B. Normality of Residuals:

- *(Histogram of Residuals)* The histogram shows deviations from a perfect bell shape, with some skewness.
- *(Q-Q Plot)* The Q-Q plot shows deviations in the tails; residuals do not align perfectly with the diagonal.

C. Constant Variance

- *(Scale-Location Plot)* The Scale-Location plot shows a fan-shaped pattern (residual spread increases as fitted values increase). Thus the residuals show non-constant variance.

9. Would it be a good idea to use the fitted linear regression equation to predict `lean` in year 2010? Explain your answer.

Answer:

No, it would not be a good idea. Based on the residual plots above, the model violates several assumptions **(i.e. Linearity, Normality of Residuals, Constance Variance)** that are critical for reliable predictions in linear regression.